

REVIT CONCRETE STRUCTURE TRAINING

Table of Contents

1. Brief overview of Revit (what it entails)	5
❖ Difference with AutoCAD software	5
❖ Benefits over AutoCAD software	5
2. Setting up the software	5
❖ Brief introduction of Libraries required	5
❖ Downloading other extensions from Autodesk site	5
3. Beginning with Revit.....	5
❖ Creating a new project	5
❖ An overview of the templates	5
❖ An overview of palettes, tabs and panels	5
4. Creating Datum Elements	6
❖ Creating Levels	6
❖ Creating grids – (same as with the levels creation)	6
❖ Propagate options extents on the grid	6
❖ Customizing and changing the shape and appearance of grid bubbles	6
❖ Linking CAD files to create grids	7
5. Brief Overview of Creating 3D elements.....	7
❖ Foundations, Columns, Beams, Slabs	7
6. Creating Columns	7
❖ Placement of columns.....	7
❖ Loading column families.....	7

7.	Creating Foundations	7
❖	Hosting foundations on the columns created	7
❖	Creating Strip footing	7
❖	Using eccentricity option	8
❖	Basement floor slab	8
❖	Loading foundation families into your drawings	8
8.	Creating Beams and slabs	8
❖	Copying beams and slabs from level to level	8
❖	Creating a slab	8
❖	Hollow pot and waffle slab	8
9.	Curved Beams	9
❖	Procedure of modeling the curved beams	9
10.	Beam System Layout and Justification Rules	9
❖	Using the beam system to model the structural beams	9
❖	Modelling the beams using the beam system layout rules	9
❖	Modelling the beams using the beam system Justification Rules	10
❖	Deselecting the beam system	10
11.	Creating Staircase and Ramps	11
❖	Types of staircase	11
❖	Creating an architectural ramp and structural ramp	11
❖	Modeling Staircase as a ramp	11
❖	Shaft openings	11
12.	Visibility/Graphics and Filters	12
❖	Applying Visibility/Graphic Overrides → Method 1	12
❖	Applying Visibility/Graphic Overrides → Method 2	12
❖	Applying Visibility/Graphic Overrides → Method 3	13
❖	Applying Visibility/Graphic Overrides → Method 4	13

13.	Creating Roofs and Truss placement.....	14
❖	Sloped and gable ended roofing	14
❖	Adding system trusses on your roof.....	14
❖	Editing and creating your own customized trusses.....	14
14.	Placing of Reinforcement	15
❖	Creating sections in your drawing	15
❖	Foundation Reinforcement	15
❖	Column Reinforcement	16
❖	Beam Reinforcement	16
❖	Slab reinforcement	16
❖	Staircase reinforcement	17
❖	Ramp reinforcement	17
15.	Annotations.....	17
❖	Adding dimensions	17
❖	Adding spot elevations	17
❖	Adding spot coordinates.....	17
❖	Adding spot slope	17
16.	Detailing of Sections	17
❖	Placing breaklines in the sections	17
❖	Using hatch files to create regions.....	18
❖	Placement of rebar tags on your reinforcements.....	18
❖	Adding comments on your reinforcement.....	18
❖	Tagging of the structural elements	18

17.	Schedules and Material Takeoffs	19
❖	Creating Bar Bending Schedules	19
❖	Loading shape images into bar bending schedules	19
❖	Creating Material Takeoffs for the structural elements.....	19
❖	Sorting and filtering fields in material takeoffs and bar bending schedules	19
18.	Placing Drawings on Sheets	19
❖	Loading different paper sizes for your title blocks	19
❖	Using system loaded title blocks	19
❖	Editing and customizing the title blocks.....	19
❖	Adjusting texts and fields in your title block	19
❖	Saving title blocks as a Revit family template	20
❖	Introduction to duplicate options for drawings.....	20
❖	Placing drawings on the sheets.....	21
19.	Printing Sheets to PDF format	21
❖	Setting up your sheets before printing	21
❖	Printing one sheet at a time	21
❖	Printing several sheets at a time	21
20.	Conversion of your files into CAD and other formats	22
❖	Export options for your drawing into other formats	22
❖	Transporting your drawing to AutoCAD.....	22
❖	Sending your model for analysis	22